Data Visualization Final Project - Chicago Violence, SVI and Social Metrics

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Introduction, Background and Research Question

Though many analyses of Chicago crime have been done, there exists a gap in existing data as it pertains to a key question: how effective are social vulnerability, mental, and developmental factors in predicting crime across Chicago? Our project examines this relationship using data from the Center for Disease Control and Prevention (CDC), the Chicago Health Atlas, and the Chicago Police Department's Citizen Law Enforcement Analysis and Reporting (CLEAR) database. We ask whether violent crime rates are systematically higher in census tracts and zip codes with greater social vulnerability, aiming to shed light on how to better target anti-crime and anti-poverty interventions, and how to best predict crime and allocate resources.

SVI and Crime in Chicago

We combined incident-level violent crime data for Chicago from 2021–2025 with census-tract SVI values and population data to examine how crime varies across geographies. After cleaning the data, we converted crime dates into a monthly time series and merged tract-level measures to enable geographic comparisons. Given that geographies have varying populations, we focused primarily on population-adjusted measures, calculating annualized and monthly violent crime rates per 1,000 residents. To analyze how vulnerability relates to crime, we grouped tracts into standard SVI quintiles and then created additional upper-tail bins (81–95, 95–98, 98–99, and 99+) to better capture variation among the most socially vulnerable areas.

Figure 1:

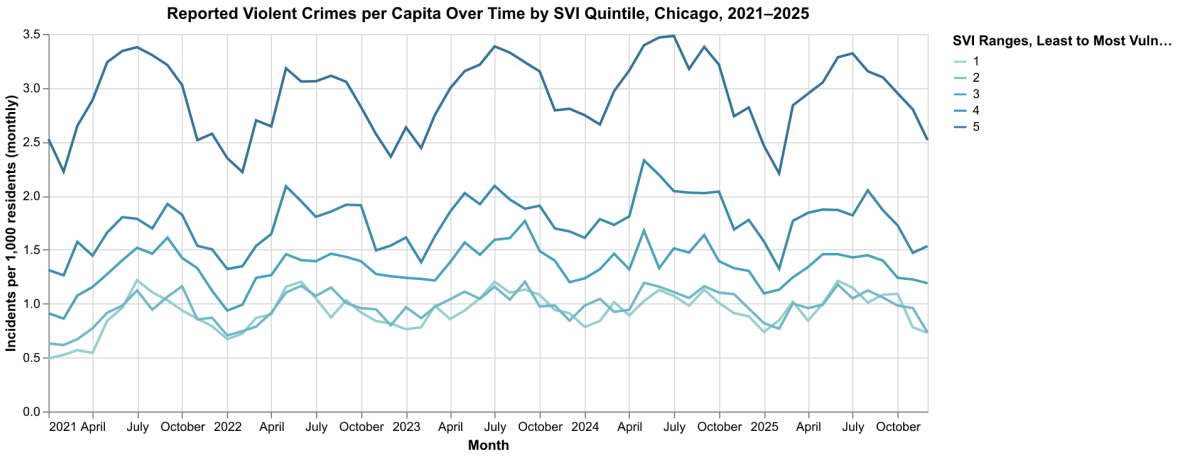
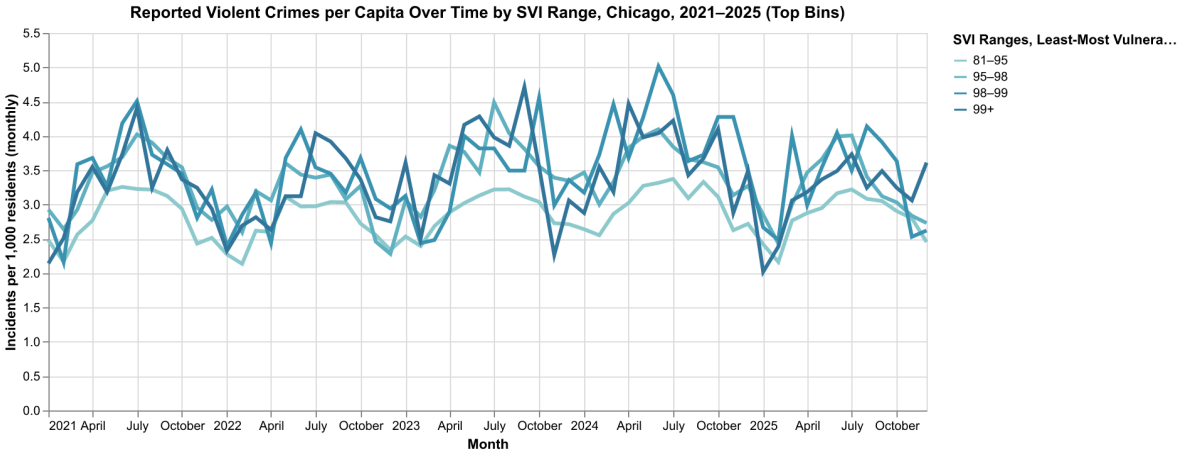
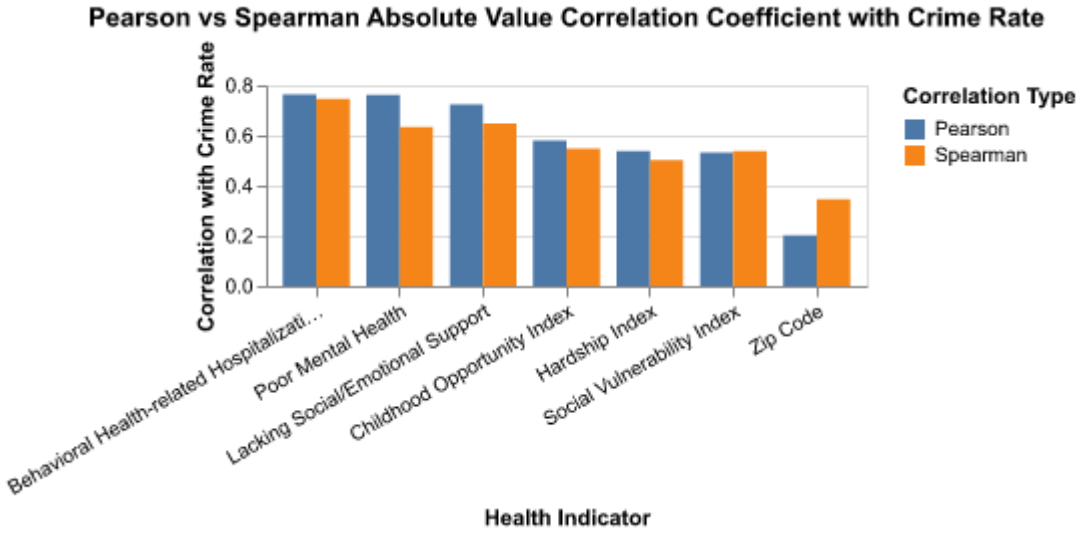


Figure 2:



Looking at figure 1, we see that violent crime incidence per capita is generally higher in more vulnerable areas. While a crime-rate-SVI correlation is not inherently surprising, the top quintile (i.e. most vulnerable) nearly doubled the crime rate of the next-most vulnerable quintile — a dramatic increase that suggests the most vulnerable are far more likely to commit violent crime. To identify if the highest SVI percentiles were the driving force behind this, we created additional bins. Looking at figure 2, we see that, while the 81st-95th percentile bin is has lower crime rates than 95+, there isn't a significant difference in crime rates among the most vulnerable 5% of tracts.



After calculating the zip code crime rate per 1,000 residents, we calculated the correlation between the crime rate and six indicators using the Pearson and Spearman's correlation coefficients, the latter being a nonparametric measure to account for potential nonlinear relationships. In our initial calculations, we found no evidence of strong correlation in any of the variables except Behavioral Health-Related Hospitalizations. Visualizing the data, we saw three zip codes in the Loop (60602, 60603, and 60604) had low SVI but crime rates over 3X almost any other zip code. Given the numerous luxury apartment spaces in this downtown area, it makes sense they would be outliers in our data, and we removed these points from our analysis to account for this.

After removing these three points, the absolute value of both the Pearson and Spearman coefficients for all our indicators were over .5, indicating moderate correlation with the crime rate. In the bar graph above, we can see that Behavioral Health-related Hospitalizations, Self-reported Mental Health and Self-reported Lacking Social/Emotional Support show the strongest correlations with a steep drop-off for the three index values. Somewhat surprisingly, this indicates that single-value statistics, including several self-reported measures, may be more accurate in predicting crime rates, compared to index values aggregating a range of statistics.

https://apprepository-s2kxjajnqkzwyg8ys7j2hm.streamlit.app/

To better understand how rates of violent crime were distributed, we created an interactive graph charting the crime rate and can be split into different zip code percentile groups. The model highlights that the top 5% of zip codes have substantially higher crime rates with the average crime rate for these zip codes almost always twice the peak values of any other percentile group.

Additionally, we built a Streamlit dashboard that allows the user to explore ZIP-code-level patterns interactively. The dashboard displays a map of Chicago ZIP codes and includes a dropdown menu for selecting a metric, such as crime per 1,000 residents, hardship, childhood opportunity, or behavioral health-related hospitalization measures. A slider highlights the ZIP code whose selected metric is closest to a chosen value, while hovering over a ZIP code reveals a richer set of contextual statistics.

Conclusion

Our findings indicate that policy efforts by the city of Chicago to address violent crime should be specific, place-based policies targeting the 5% of zip codes with substantially higher rates of violent crime than the rest of the city. Additionally, we found that single-statistic predictors and several self-reported mental health factors were strong predictors for the crime rate, exceeding more comprehensive index values.